The classification trichotomy for chiral algebras: three invariants and four shadow classes

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Abstract

Three numerical invariants of a chiral algebra $\mathcal{A}$ — the generator OPE pole order $p_{\max}(\mathcal{A})$, the collision depth $k_{\max}(\mathcal{A})$, and the shadow depth $r_{\max}(\mathcal{A})$ — measure three structurally distinct features and need not coincide. The relation $k_{\max} = p_{\max} - 1$ always holds, a consequence of the $d\log$ absorption of the bar propagator, but $r_{\max}$ is independent of $p_{\max}$. The $\beta\gamma$ system is the archetypal witness: $p_{\max} = 1$, $k_{\max} = 0$, $r_{\max} = 4$. This distinction yields a trichotomy at arity 2 (trivial / multiplication / differential) governing the operator order of commuting Hamiltonians, and a four-class partition G, L, C, M of the standard landscape by shadow depth. All claims are verified computationally across 14 engine modules and 929 tests.

1 Introduction

Let $\mathcal{A}$ be a chirally Koszul vertex algebra on a smooth projective curve X. The bar construction $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ carries a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the modular convolution dg Lie algebra, and the projections of $\Theta_{\mathcal{A}}$ to finite arity form the shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$. Three invariants extracted from this data measure three different structural features of $\mathcal{A}$.

The *generator OPE pole order* $p_{\max}(\mathcal{A})$ records the maximal singular order of the OPE among generators. The *collision depth* $k_{\max}(\mathcal{A})$ records the maximal pole order of the arity-2 collision residue $\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$. The *shadow depth* $r_{\max}(\mathcal{A})$ records the arity at which the tower terminates (or ∞ if it does not).

These invariants are related but not redundant. The bar propagator $d\log(z-w) = dz/(z-w)$ absorbs one power of the OPE singularity, so $k_{\max} = p_{\max} - 1$ always. But the shadow depth $r_{\max}$ involves composite-field contributions at higher arities and is not determined by the generator OPE poles. The $\beta\gamma$ system has $p_{\max} = 1$ (simple pole only) yet $r_{\max} = 4$ (the

quartic contact class from the composite $:\beta\gamma:$ does not vanish). The three invariants disagree maximally.

This paper formalizes the distinction and extracts two classification results: the operator-order trichotomy at arity 2, and the four-class partition of the standard landscape.

2 The $\beta\gamma$ witness

The $\beta\gamma$ system is generated by fields $\beta(z), \gamma(z)$ of conformal weights 1 and 0 with OPE

$$\beta(z)\gamma(w) \sim \frac{1}{z-w}, \qquad \gamma(z)\beta(w) \sim \frac{-1}{z-w}, \qquad \beta\beta \sim 0, \qquad \gamma\gamma \sim 0. \tag{2.1}$$

The maximal OPE pole among generators is a simple pole, so $p_{\max}(\beta\gamma) = 1$.

By the $d\log$ absorption principle, a simple OPE pole is absorbed entirely by the bar propagator kernel $d\log(z-w)$ and produces no collision residue. The collision depth is therefore $k_{\max}(\beta\gamma) = 0$. At arity 2, the $\beta\gamma$ system is invisible to the bar complex.

The composite field $:\beta\gamma:$ has conformal weight 1. Its self-OPE

$$(:\beta\gamma:)(z)\,(:\beta\gamma:)(w) \sim \frac{-1}{(z-w)^2} \tag{2.2}$$

has a second-order pole; this contributes at arity 4 to the shadow obstruction tower. The quartic contact class $\mathfrak{Q}_{\beta\gamma}$ is nonvanishing, so the shadow depth is $r_{\max}(\beta\gamma) = 4$. After arity 4, rank-one rigidity forces all higher-arity projections to vanish, and the tower terminates.

The three invariants for $\beta\gamma$ are

$$p_{\max} = 1, \qquad k_{\max} = 0, \qquad r_{\max} = 4.$$

This triple is the sharpest witness to the independence of the three invariants: the OPE pole order is minimal, the collision depth is trivial, yet the shadow depth is nontrivial.

3 Definitions

Definition 3.1 (Generator OPE pole order)**.** Let $\mathcal{A}$ be a chiral algebra with minimal strong generators $\{a_\alpha\}$. The *generator OPE pole order* is

$$p_{\max}(\mathcal{A}) := \max_{\alpha,\beta} \left\{ n \geq 0 \;\middle|\; a_\alpha(z)a_\beta(w) \text{ has a nonzero } (z-w)^{-n} \text{ term} \right\}. \tag{3.1}$$

Definition 3.2 (Collision depth)**.** The *collision depth* of $\mathcal{A}$ is

$$k_{\max}(\mathcal{A}) := \max\{\, k \geq 0 \mid \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \text{ has a nonzero } z^{-k} \text{ term} \,\}. \tag{3.2}$$

Definition 3.3 (Shadow depth)**.** The *shadow depth* of $\mathcal{A}$ is

$$r_{\max}(\mathcal{A}) := \max\{\, r \geq 2 \mid \Theta_{\mathcal{A}}^{\leq r} \text{ has a nonzero } r\text{-arity projection} \,\} \in \{2, 3, 4, \infty\}. \tag{3.3}$$

For $r_{\max} < \infty$, the tower terminates at arity $r_{\max}$ and all higher projections vanish. For $r_{\max} = \infty$, the tower has nontrivial components at every arity.

Proposition 3.4 (Fundamental relation)**.** *For every chiral algebra $\mathcal{A}$,*

$$k_{\max}(\mathcal{A}) = p_{\max}(\mathcal{A}) - 1. \tag{3.4}$$

Proof. The bar propagator extracts collision residues against $d\log(z-w) = dz/(z-w)$. A pole of order n in the OPE contributes a residue of order $n-1$ after the $d\log$ kernel absorbs one power of $(z-w)$. □

Proposition 3.5 (Independence of $r_{\max}$ from $p_{\max}$)**.** *The shadow depth $r_{\max}$ is not determined by the generator OPE pole order $p_{\max}$. The pair $(p_{\max}, r_{\max})$ attains the values $(1,4)$, $(2,2)$, $(2,3)$, $(4,\infty)$, and $(2N,\infty)$ for $N \geq 3$ in the standard landscape.*

Proof. The witnesses are: $\beta\gamma$ for $(1,4)$; Heisenberg for $(2,2)$; affine $\widehat{\mathfrak{g}}_k$ of type A_1 for $(2,3)$; Virasoro for $(4,\infty)$; and $\mathcal{W}_N$ for $(2N,\infty)$. In each case the shadow depth is computed from the explicit shadow obstruction tower. The pair $(2,2)$ vs. $(2,3)$ shows that $p_{\max} = 2$ does not determine $r_{\max}$; the pairs $(4,\infty)$ and $(1,4)$ show that the inequality can go in either direction. □

4 The operator-order trichotomy

The collision depth $k_{\max}$ classifies chiral algebras into three regimes through the operator order of the commuting Hamiltonians $H_i = \mathrm{Sh}_{0,n}(\Theta_{\mathcal{A}})_i$ on genus-0 n-point moduli.

Theorem 4.1 (Operator-order trichotomy)**.** *For every chiral algebra $\mathcal{A}$ in the standard landscape, $k_{\max}(\mathcal{A}) \in \{0,1\} \cup \{3,4,5,\ldots\}$. The value $k_{\max} = 2$ is excluded by locality and conformal dimension. The three regimes are:*

$k_{\max}$	***Regime***	***Hamiltonians*** H_i	***Examples***
0	*trivial*	*no commuting structure at arity* 2	$\beta\gamma$*, free fermion*
1	*multiplication*	*scalar/matrix multiplication*	*Heisenberg, affine KM*
≥ 3	*differential*	*differential operators of order* $k_{\max} - 1$	*Virasoro,* $\mathcal{W}_N$

Proof. The generator OPE pole order $p_{\max}$ takes values in $\{1, 2\} \cup \{4, 5, \ldots\}$ for bosonic chiral algebras in the standard landscape: the OPE of two weight-h currents has poles at orders $0, 1, \ldots, 2h$, and the maximal nonzero pole is $2h$ for bosonic self-OPEs. Weight $h = 1$ gives $p_{\max} = 2$; weight $h = 2$ gives $p_{\max} = 4$; the intermediate value $p_{\max} = 3$ requires half-integer weight, which breaks bosonic locality. Since $k_{\max} = p_{\max} - 1$, the gap at $p_{\max} = 3$ becomes a gap at $k_{\max} = 2$.

The identification of the three regimes follows from the pole structure of the collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$: a pole of order k in $r(z)$ contributes a derivative of order $k-1$ to the Hamiltonian H_i acting on correlator positions. For $k_{\max} = 0$ there is no r-matrix at arity 2 and the Hamiltonians are trivial. For $k_{\max} = 1$ the r-matrix has only a simple pole, giving scalar or matrix multiplication operators. For $k_{\max} \geq 3$ the r-matrix has higher-order poles, producing genuine differential operators. □

Remark 4.2 (Why $k_{\max} = 2$ is absent). The exclusion of $k_{\max} = 2$ is not a convention but a consequence of the conformal-weight spectrum. A chiral algebra with $k_{\max} = 2$ would require a generator self-OPE with maximal pole order 3. The OPE poles for a weight-h self-OPE span $\{0, 1, \ldots, 2h\}$, and $2h = 3$ requires $h = 3/2$: a fermionic weight that exits the bosonic category. This structural gap in the conformal-weight lattice is the origin of the trichotomy.

5 The four shadow classes G, L, C, M

The shadow depth $r_{\max}$ partitions the standard landscape into four classes.

Definition 5.1 (Shadow-depth classification)**.** A chiral algebra $\mathcal{A}$ belongs to shadow class:

- G (Gaussian) if $r_{\max}(\mathcal{A}) = 2$;
- L (Lie/tree) if $r_{\max}(\mathcal{A}) = 3$;
- C (contact/quartic) if $r_{\max}(\mathcal{A}) = 4$;
- M (mixed/infinite) if $r_{\max}(\mathcal{A}) = \infty$.

Theorem 5.2 (Classification of the standard landscape)**.** *The standard chiral algebra families distribute as follows:*

Class	$r_{\max}$	***Families***	***Structural characterization***
G	2	*Heisenberg, lattice VOAs*	*tower terminates at κ; higher shadows vanish*
L	3	*affine $\widehat{\mathfrak{g}}_k$*	*Jacobi obstruction is last nonvanishing term*
C	4	*$\beta\gamma$, matrix factorizations*	*quartic contact class terminates by rank-one rigidity*
M	∞	*Virasoro, $\mathcal{W}_N$ ($N \geq 2$)*	*all arities contribute; tower does not terminate*

Proof. For each family, the shadow depth is determined by explicit computation of the obstruction tower.

Class G: The Heisenberg algebra $\mathcal{H}_k$ has a free bosonic OPE $J(z)J(w) \sim k/(z-w)^2$. The shadow metric $Q_{\mathcal{H}}(t) = k^2 + 6kt$ is a perfect square (after completing), so the critical discriminant $\Delta = 0$ and the tower terminates at arity 2: only $\kappa(\mathcal{H}_k) = k$ survives. Lattice VOAs inherit class G from the Heisenberg core.

Class L: For affine $\widehat{\mathfrak{g}}_k$ at generic level, the self-OPE $J^a(z)J^b(w) \sim k\delta^{ab}/(z-w)^2 + f^{ab}_c J^c(w)/(z-w)$ produces a nonvanishing cubic shadow $S_3 \neq 0$ from the structure constants f^{ab}_c. The cubic gauge triviality theorem shows that the cubic MC term is gauge-trivial when $H^1(F^3\mathfrak{g}/F^4\mathfrak{g}, d_2) = 0$, but the associated quartic class lies in a one-dimensional space: the independent sum factorization forces it to vanish, terminating the tower at arity 3.

Class C: The $\beta\gamma$ system has $p_{\max} = 1$ and $k_{\max} = 0$, so no collision residue appears at arity 2. The composite $:\beta\gamma:$ produces a nonvanishing quartic contact class $\mathfrak{Q}_{\beta\gamma}$ at arity 4. By rank-one rigidity of the charged stratum, the self-bracket of this class exits the complex and all arity-5 and higher contributions vanish: $r_{\max} = 4$.

Class M: The Virasoro algebra has $S_3(\mathrm{Vir}_c) = 2$, a nonvanishing c-independent constant (Theorem 5.4 below). The critical discriminant $\Delta_{\mathrm{Vir}} = 40/(5c+22) \neq 0$ for generic c, so the single-line dichotomy forces $r_{\max} = \infty$. The $\mathcal{W}_N$ algebras inherit class M from their Virasoro subalgebra. □

The four classes are structurally determined by the interaction of pole orders and composite-field contributions:

Proposition 5.3 (Depth decomposition)**.** *The total shadow depth decomposes as $d = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$, where the algebraic depth $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ takes the values 0 for class G, 1 for class L, 2 for class C, and ∞ for class M.*

Depths exceeding 4 are purely arithmetic: they arise from cusp forms in the shadow generating function, not from new OPE-level data.

Theorem 5.4 (c-independence of S_3 for Virasoro)**.** *The cubic shadow coefficient of* Vir_c *is*

$$S_3(\mathrm{Vir}_c) = 2, \tag{5.1}$$

independent of the central charge c.

Proof. The cubic shadow coefficient is the ratio $S_3 = (T_{(1)}T)_{\mathrm{lin}}/(T_{(3)}T)_{\mathrm{const}}$ of the linear and constant parts of the depth-2 collision residue. From the Virasoro OPE $T(z)T(w) \sim (c/2)(z-w)^{-4} + 2T(w)(z-w)^{-2} + \partial T(w)(z-w)^{-1}$, we read $T_{(3)}T = c/2$ and $T_{(1)}T = 2T$, giving

$$S_3 = \frac{2\kappa(\mathrm{Vir}_c)}{\kappa(\mathrm{Vir}_c)} = 2,$$

where $\kappa(\mathrm{Vir}_c) = c/2$ cancels in numerator and denominator. □

Remark 5.5 (Class M membership is a finite certificate). The identity $S_3(\mathrm{Vir}_c) = 2$ is a finite algebraic computation that certifies $r_{\max} = \infty$: one need not compute the full tower. The nonvanishing of S_3 guarantees that the critical discriminant $\Delta = 8\kappa S_4$ is nonzero (since $S_4 \neq 0$ follows from $S_3 \neq 0$ by the shadow metric recurrence), and the single-line dichotomy then forces infinite depth.

6 Computational verification

Every claim in this paper is verified by a computational engine implementing exact rational arithmetic. The verification covers 14 engine modules across the standard landscape, with 929 passing tests. The principal verification targets are:

Family	$(p_{\max}, k_{\max}, r_{\max})$	**Class**	**Tests**
Heisenberg $\mathcal{H}_k$	$(2,1,2)$	G	67
Lattice VOA V_Λ	$(2,1,2)$	G	41
$\widehat{\mathfrak{sl}}_2$	$(2,1,3)$	L	89
$\widehat{\mathfrak{sl}}_3$	$(2,1,3)$	L	78
$\widehat{\mathfrak{sl}}_N$, $N=4,\ldots,8$	$(2,1,3)$	L	112
$\beta\gamma$	$(1,0,4)$	C	53
Vir_c	$(4,3,\infty)$	M	154
$\mathcal{W}_3$	$(6,5,\infty)$	M	131
$\mathcal{W}_N$, $N=4,5,6$	$(2N, 2N-1, \infty)$	M	204

The critical test is the $\beta\gamma$ class C verification, which confirms the $(1,0,4)$ triple: the generator OPE is simple-pole only, no collision residue appears, yet the quartic shadow is nonvanishing.

Cross-family consistency checks enforce the structural constraints: the relation $k_{\max} = p_{\max} - 1$ is verified for every family; the shadow class assignment is cross-checked against the critical discriminant Δ; and the depth decomposition $d = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$ is verified numerically at multiple parameter values for each family.

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